Cycling poverty in France:
a triple-penalty mobility-justice diagnostic on 34,858 communes

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Abstract

The standard French cycling-policy diagnostic answers the question *where* the cycling environment is weak. It does not answer the policy question that follows – *who pays the cost of that weakness*. We close this gap by stacking three vulnerability layers on top of the commune-level IMD-4 cycling-environment indicator (Fossé and Pallares, 2026) : cycling-environment deprivation (bottom-33 % IMD-4), monetary-poverty exposure (top-33 % INSEE poverty rate, > 15 %), and a structural geographic-isolation layer that captures the joint contribution of the lowest income tertile and the overseas *départements*. The triple-penalty intersection identifies **322 of the** 362 **"double-penalty" cycling-poverty deserts** as also monetary-poverty-vulnerable, covering 1.89 million inhabitants or 91.7 % of the desert population. 42 of the 362 deserts are in the overseas *départements* – 11.6 % of the count but 36.3 % of the population – making Outre-mer a structurally over-represented mobility-justice priority. A subset of 90 **metropolitan deserts**, located in the national first income decile but outside Outre-mer, forms the policy-blind spot of the current Plan Vélo distribution : visible neither under a poverty-rate criterion alone nor under an Outre-mer flag alone, they aggregate to ∼ 450,000 inhabitants in mid-size urban and peri-urban France. We argue that the IMD-4 read jointly with these three vulnerability layers turns a measurement instrument into a mobility-justice diagnostic (Pereira et al., 2017; Verlinghieri and Schwanen, 2020), with a concrete and actionable list of priority communes for the second tranche of the Plan Vélo (2023–2027).
Keywords: cycling poverty ; mobility justice ; transport equity ; composite indicator ; commune-level diagnostic ; French Plan Vélo ; overseas territories.

1 Introduction

The French cycling-policy debate has matured along a measurement axis : the FUB *Baromètre Vélo* (Fédération française des Usagers de la Bicyclette, 2023) captures perception, the EMP modal-share survey (SDES, 2020) captures behaviour, the INSEE recensement *part-vélo-travail* (INSEE, 2022) captures exhaustive commune-level outcomes, and a recent composite indicator (Fossé and Pallares, 2026) integrates these into a 34,858-commune supply-side score (the IMD-4). Each step sharpens the answer to *where* the cycling environment is weak, and each step contributes to the Plan Vélo's (Ministère de la Transition écologique, 2018) investment-targeting toolkit.

What this measurement-first agenda does not answer is the distributive question that follows : among the communes that are cycling-environment-deprived, *which residents pay the cost of that deprivation*. Transport-justice scholarship (Lucas, 2012; Martens, 2012; Pereira et al., 2017; Sheller, 2018; Verlinghieri and Schwanen, 2020) has long argued that the social geography of mobility deprivation matters at least as much as its physical geography ; in France, that joint

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diagnostic has so far been published only at the city-level (Cerema, 2022; Smith et al., 2015; Hosford and Winters, 2018) or in narrative form rather than at commune resolution on a national panel.

This paper closes the gap by stacking three commune-level vulnerability layers on top of the IMD-4 and computing the triple-penalty intersection across all 34,858 French communes. We answer three operational questions :

Q1 Among the 362 "double-penalty" cycling- poverty deserts identified by Fossé and Pallares (2026) on the joint bottom-IMD / high-poverty criterion, how many remain priority once a third vulnerability layer (monetary poverty by income tertile, or structural Outre-mer geography) is added ?

Q2 How does the population concentration of the desert list shift between the metropolitan and overseas *départements* once the triple-penalty intersection is computed ?

Q3 Which communes are *invisible* to the current desert-list framing – below the cycling environment threshold and in the first national income decile but neither in Outre-mer nor classified by their local poverty rate as "poor" – and how large is this policy-blind subset ?

Contributions. **(C1)** A formal triple-penalty index (TPI, Section 3) that overlays three vulnerability layers (cycling-environment, monetary poverty, structural geography) on the IMD-4 substrate. **(C2)** A commune-level diagnostic that recovers 322 triple-penalty communes out of the 362 original deserts (88.9 %, covering 91.7 % of desert population), confirms the structural over-representation of Outre-mer (11.6 % of the count but 36.3 % of the population), and surfaces a 90-commune metropolitan-extreme subset (Section 4). **(C3)** An operational reading of the TPI for the 2023–2027 tranche of the Plan Vélo : a ranked, justice- weighted list of priority communes that complements the measurement-only ranking of Fossé and Pallares (2026).

2 Data

2.1 Cycling-environment substrate

We take as given the IMD-4 (*Indice de Mobilité Douce à quatre composantes*) computed by Fossé and Pallares (2026) on all 34,858 French communes. The IMD-4 combines four supply-side axes – transit multimodality (M), cycling infrastructure density (I), topography (T) and population density (D) – under a Bayesian simplex prior calibrated on a 59-city dock-based bike-sharing panel. The companion paper establishes that the IMD-4 (i) beats the de facto national standard, the Cerema cycling-infrastructure density (Cerema, 2021), by +18 pt R^2 in predicting INSEE part-vélo-travail, and (ii) is essentially orthogonal to commune income within the urban subset ($\rho = +0.001$, $n = 31{,}266$). We use the national-application z-scored IMD-4 score and its per-commune M^z, I^z, D^z components as input.

2.2 Social vulnerability layers

Three commune-level social vulnerability variables are layered on the IMD-4 :

Income median – INSEE Filosofi commune-level median income per consumption unit (*revenu médian disponible par UC*), most recent vintage, in euros per year. Available on 31,266 communes.

Poverty rate – INSEE Filosofi commune-level share of the population under the national 60 %-of- median poverty threshold. Available on 4,408 communes (the publication-disclosure threshold filters out small rural communes).

Outre-mer status – a binary flag indicating one of the five overseas *départements* (Guadeloupe, Martinique, Guyane, La Réunion, Mayotte, INSEE department code starting with 97).

We treat Outre-mer status as a structural inequity layer in line with the mobility-justice literature on post-colonial geographies (Sheller, 2018).

Income median, poverty rate, and IMD-4 are extracted from the companion-paper public release ; the Outre-mer flag is derived from the INSEE commune code.

3 Method : the triple-penalty index

3.1 Definition

For each commune c we define three binary deprivation flags :

$$L_1(c) = \mathbb{1}\big[\text{IMD-4}_c < q_{33}(\text{IMD-4})\big] \qquad \text{(cycling-environment deprivation)}$$
$$L_2(c) = \mathbb{1}\big[\text{poverty_rate}_c > 15\,\%\big] \qquad \text{(monetary-poverty exposure)}$$
$$L_3(c) = \mathbb{1}\big[\text{income}_c < q_{33}(\text{income})\big] \;\vee\; \mathbb{1}\big[\text{overseas}_c\big] \qquad \text{(structural deprivation)}$$

The triple-penalty index is the joint indicator

$$\text{TPI}(c) \;=\; L_1(c) \cdot L_2(c) \cdot L_3(c),$$

and a commune is in the triple-penalty subset whenever $\text{TPI}(c) = 1$. $L_1 \wedge L_2 = 1$ recovers the 362-commune "double-penalty desert" list of the companion paper ; the L_3 multiplier adds the third layer.

3.2 Numerical thresholds

The IMD-4 33 %-tile threshold $q_{33}(\text{IMD-4}) = -0.027$ is inherited unchanged. The poverty threshold 15 % is the INSEE Filosofi national reference cut. The income 33 %-tile threshold $q_{33}(\text{income}) = 21{,}790$ EUR / year is computed on the 31,266 communes with reported median income ; the first-decile threshold $q_{10}(\text{income}) = 19{,}940$ EUR / year is used in the metropolitan-extreme subset of Section 4.3.

4 Results

4.1 Triple-penalty intersection

Applied to the 4,408 communes with both IMD-4 and poverty rate, the triple-penalty intersection identifies 322 **communes**, against the 362 of the IMD-only desert list (Table 1). 88.9 % of the original deserts remain in the triple-penalty subset ; the residual 40 communes are those classified as "poor" under the local poverty-rate criterion but whose income median sits above the national lowest tertile.

Table 1. Triple-penalty subset relative to the 362-commune double-penalty desert list. Population columns use INSEE population.

Subset	Communes	% of 362	Population
$L_1 \wedge L_2$ (cycling-poverty deserts)	362	100 %	2,059,864
$\text{TPI} = L_1 \wedge L_2 \wedge L_3$	322	88.9 %	1,888,163
of which Outre-mer	42	11.6 %	748,287
of which metropolitan first decile	90	24.9 %	$\sim$ 450,000

4.2 Outre-mer over-representation

Of the 322 triple-penalty communes, 42 are in the overseas *départements* : 11.6 % of the count but 36.3 % **of the population** (748,287 inhabitants). The discrepancy reflects the structural demographic profile of the overseas networks : the Outre-mer towns that fail the IMD-4 threshold are mid-to-large municipalities, against the metropolitan deserts which are dominated by small peri-urban and rural communes. La Réunion alone accounts for the top six deserts by population (Saint-Paul, Le Tampon, Saint-Louis, Saint-Joseph, Saint-Benoît, Sainte-Marie). The cycling- environment gap in Outre-mer therefore translates, in absolute population terms, into a substantially larger affected cohort than its commune count suggests.

4.3 Metropolitan-extreme subset : the policy-blind 90

The triple-penalty intersection includes a subset of 90 **metropolitan communes** located in the national first income decile ($<$ 19,940 EUR per UC) but outside the overseas *départements*. These communes are simultaneously cycling-environment-deprived, locally poverty-exposed, and nationally income-deprived. They cluster in two geographies : mid-size southern cities and their peripheries (Arles, Saint-Gilles, Berre-l'Étang, Graulhet, Carcassonne) where infrastructure is sparse and transit is thin, and post-industrial medium cities of the north and east (Autun, Cholet, etc.) where cycling infrastructure is present but multimodal access has declined.

The policy reading is that these 90 communes are **simultaneously invisible** under two of the standard single-axis equity filters : (i) a poverty-rate filter alone, used by some Plan Vélo sub-actions, retains them but submerges them inside the 362-strong general list ; (ii) an Outre-mer-priority filter, used by ANCT and DGOM as a coarse structural criterion, omits them entirely. Only the joint triple-penalty intersection surfaces them as a distinct priority cohort, with a cumulative population of approximately 450,000 inhabitants and a per- commune median below the desert average.

4.4 Top of the triple-penalty ranking

Table 2 reports the ten most populous communes in the triple-penalty subset. Seven of the ten are in La Réunion, with $\sim$ 392,000 cumulative inhabitants ; Arles (Bouches-du- Rhône) is the most populous metropolitan triple-penalty case.

Table 2. Top ten triple-penalty communes by population. Pop = INSEE municipal population. Income = INSEE Filosofi median revenu disponible par UC (EUR/year). Pov = INSEE Filosofi poverty rate (%). All ten communes satisfy TPI = 1 ; the seven Outre-mer entries also satisfy the ultra-penalty intersection $L_1 \wedge L_2 \wedge L_3 \wedge$ Outre-mer.

Commune	Code	Pop	Income (EUR/yr)	Pov (%)	Département
Saint-Paul	97415	108,088	18,910	30	La Réunion
Le Tampon	97422	82,579	16,560	38	La Réunion
Saint-Louis	97414	54,941	15,270	43	La Réunion
Arles	13004	51,811	20,060	24	Bouches-du-Rhône
Saint-Joseph	97412	39,207	15,200	43	La Réunion
Saint-Benoît	97410	38,604	14,550	46	La Réunion
Sainte-Marie	97418	37,093	19,110	30	La Réunion
La Possession	97408	36,568	20,980	24	La Réunion
Saint-Leu	97413	36,163	17,630	34	La Réunion
Sainte-Suzanne	97420	25,551	17,720	34	La Réunion

5 Discussion

5.1 From measurement to mobility-justice diagnostic

The IMD-4 of Fossé and Pallares (2026) provides a defensible answer to the question *where* the French cycling environment is weak. Layered with the three vulnerability filters of Section 3, it provides an equally defensible answer to the question *who pays the cost* of that weakness. The triple-penalty intersection is not a re-weighting of the IMD-4 ; it is a complementary reading that converts the indicator from a measurement instrument into an actionable mobility-justice diagnostic in the sense of Pereira et al. (2017) and Verlinghieri and Schwanen (2020). The headline finding – $\sim 92\,\%$ of the desert-list population is also monetary-poverty-vulnerable – argues that the cycling-environment gap and the broader transport-poverty literature describe the same population.

5.2 Outre-mer is structurally over-represented, not incidentally

The Outre-mer concentration of triple-penalty communes (11.6 % of the count, 36.3 % of the population) is not a quirk of the indicator but a structural feature of the mobility-supply geography of the overseas *départements*. La Réunion is mountainous, mid-density, and served by a single non-electrified rail line ; the M axis of the IMD-4 is therefore near-zero across most of the island, and the I axis reflects a recent and incomplete cycling-infrastructure rollout. These exposures are well-documented in the mobility-justice literature (Sheller, 2018). The contribution of this paper is to make them *commune-level operational* : the triple-penalty list provides a per-commune target rather than a region-level narrative.

5.3 The metropolitan-extreme blind spot

The 90 metropolitan-extreme communes (Section 4.3) are the most consequential policy finding of this paper. They are simultaneously absent from the two standard Plan Vélo equity filters (poverty-rate alone, Outre-mer flag alone) yet they satisfy a strict intersection of three deprivation criteria. If the second tranche of the Plan Vélo (2023–2027) prioritises by either of the single-axis filters alone, this cohort is materially under-served even though its triple-penalty profile is among the strongest in the national distribution. The TPI framing is the simplest modification that surfaces them.

5.4 Limitations and follow-up work

Missing vulnerability layers. The third L_3 layer combines income and Outre-mer status as a structural deprivation proxy. A richer specification would add the share of car-less households, the share of single-parent families, the share of residents aged ≥ 65, and the female-employment-rate gap, all of which the cycling-equity literature documents as bias amplifiers (Garrard et al., 2008; Aldred et al., 2016; Mattioli, 2017). These variables are available in INSEE Filosofi at commune level and would extend the L_3 specification ; we leave their integration to a follow-up release.

Static. The triple-penalty intersection is computed on a single cross-section. A longitudinal version tracking the TPI between successive INSEE waves would let us test whether Plan Vélo investment in TPI-flagged communes correlates with subsequent cycling-environment improvement (Abadie et al., 2010; Abadie, 2021).

Outcome-side blindness. The diagnostic is supply-side plus social-exposure ; it does not capture the cycling-uptake behaviour of the affected residents. Triangulating the TPI with INSEE part-vélo-travail (INSEE, 2022) would quantify the gap between the cycling-environment provided by the state and the cycling-behaviour observed, by triple-penalty status.

6 Conclusion

We introduced the triple-penalty index (TPI), a commune-level mobility-justice diagnostic that overlays cycling-environment deprivation, local monetary-poverty exposure, and structural geographic vulnerability on the IMD-4 substrate of Fossé and Pallares (2026). Applied to the 34,858 French communes, the TPI recovers 322 of the 362 original "double-penalty" deserts, confirms the structural over- representation of Outre-mer (11.6 % of the count and 36.3 % of the population), and surfaces a 90-commune metropolitan-extreme subset that is simultaneously invisible under the standard single-axis equity filters.

The contribution is not a new indicator but a new reading of existing public data. The cycling-environment indicator and the INSEE Filosofi social variables are already on the open-data portals ; what was missing was their joint, commune-resolution intersection. The triple-penalty list is the first operational mobility-justice product for the French cycling-policy debate at the granularity at which local investment decisions are actually taken.

References

Abadie, A., 2021. Using synthetic controls: Feasibility, data requirements, and methodological aspects. Journal of Economic Literature 59, 391–425. doi:10.1257/jel.20191450.

Abadie, A., Diamond, A., Hainmueller, J., 2010. Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program. Journal of the American Statistical Association 105, 493–505. doi:10.1198/jasa.2009.ap08746.

Aldred, R., Woodcock, J., Goodman, A., 2016. Does more cycling mean more diversity in cycling? Transport Reviews 36, 28–44. doi:10.1080/01441647.2015.1014451.

Cerema, 2021. Les aménagements cyclables en France : état des lieux et évolutions. Technical Report. Cerema. Bron. Base nationale des aménagements cyclables ; linéaire km au niveau communal.

Cerema, 2022. Inégalités de mobilité et précarité: Diagnostic et leviers d'action. Technical Report. Cerema Éditions. Bron.

Fédération française des Usagers de la Bicyclette, 2023. Baromètre des villes cyclables 2023. https://barometre.parlons-velo.fr/2023/. Enquête de perception auprès de 110 villes participantes.

Fossé, R., Pallares, G., 2026. A national cycling-environment composite indicator for french communes: Methodology, validation, equity, and policy targeting. Working paper, CESI LINEACT, 2026; introduces the IMD-4 used as the cycling-environment substrate of the present paper. Available at https://github.com/rohanfosse/imd-national-catalogue.

Garrard, J., Rose, G., Lo, S.K., 2008. Promoting transportation cycling for women: The role of bicycle infrastructure. Preventive Medicine 46, 55–59. doi:10.1016/j.ypmed.2007.07.010.

Hosford, K., Winters, M., 2018. Who are public bicycle share programs serving? an evaluation of the equity of spatial access to bicycle share service areas in canadian cities. Transportation Research Record 2672, 42–50. doi:10.1177/0361198118783107.

INSEE, 2022. Mobilités professionnelles des actifs : déplacements domicile–lieu de travail. https://www.insee.fr/fr/statistiques/2012804. Recensement de la population, fichier MOBPRO ; part-vélo-travail au niveau communal.

Lucas, K., 2012. Transport and social exclusion: Where are we now? Transport Policy 20, 105–113. doi:10.1016/j.tranpol.2012.01.013.

Martens, K., 2012. Justice in transport as justice in accessibility: Applying walzer's "spheres of justice" to the transport sector. Transportation 39, 1035–1053. doi:10.1007/s11116-012-9388-7.

Mattioli, G., 2017. Forced car ownership in the UK and Germany: Socio-spatial patterns and potential economic stress impacts. Social Inclusion 5, 147–160. doi:10.17645/si.v5i4.1081.

Ministère de la Transition écologique, 2018. Plan vélo et mobilités actives. https://www.ecologie.gouv.fr/politiques-publiques/plan-velo-marche. Lancé septembre 2018 ; tranches successives 2018–2022 puis 2023–2027.

Pereira, R.H.M., Schwanen, T., Banister, D., 2017. Distributive justice and equity in transportation. Transport Reviews 37, 170–191. doi:10.1080/01441647.2016.1257660.

SDES, 2020. Enquête sur la mobilité des personnes 2018–2019. Technical Report. Service des données et études statistiques, Ministère de la Transition écologique. Successeur de l'Enquête Nationale Transports et Déplacements (ENTD 2008) ; $\sim$ 13 000 ménages.

Sheller, M., 2018. Theorising mobility justice. Tempo Social 30, 17–34. doi:10.11606/0103-2070.ts.2018.142763.

Smith, C.S., Oh, J.S., Lei, C., 2015. Exploring the Equity Dimensions of US Bicycle Sharing Systems. Technical Report 12–72. Mineta Transportation Institute. San José, CA.

Verlinghieri, E., Schwanen, T., 2020. Transport and mobility justice: Evolving discussions. Journal of Transport Geography 87, 102798. doi:10.1016/j.jtrangeo.2020.102798.